

1. a) let, no. of cycles = n

So, 3 bytes are transferred $\frac{n}{2}$ cycles
and 5 " " " " $\frac{n}{2}$ "

ATQ,

$$3 \times \frac{n}{2} + 5 \times \frac{n}{2} = 600$$

$$\Rightarrow n = \frac{600 \times 2}{8} = 150$$

Total time =

$$\begin{aligned} & \left(600 + \frac{3 \times 10^9}{40 \times 2^{10}} + 500 \right) \times \frac{150}{2} \\ & + \left(600 + \frac{5 \times 10^9}{40 \times 2^{10}} + 500 \right) \times \frac{150}{2} \\ & = 14813437.5 \text{ ns} \approx 0.0148 \text{ s} \end{aligned}$$

b) Total memory capacity = $2^8 = 256$

Reserved bit = 3, Can be used = 5 bit

$\therefore$ Current memory = 2^5

$\therefore$ Memory Reduced = $2^8 - 2^5 = 224$

So, $(224/256) \times 100\% \approx 87.5\%$ waste.

2) a) i) BX: CFB1 $\xrightarrow{0001}$
 + AX: ABCD $\xrightarrow{1101}$ = 1110; So no carry from 3rd bit to 4th bit.
 Result: 17B7E
 16 bit $\Rightarrow$ BX
 $\boxed{CF=1}$ $\therefore \boxed{AF=0}$

ii) CX: F412 $\xrightarrow{1110(-)}$
 $\xrightarrow{0111(+)}$
 - BX: 7B7E
 Result: 6894 = 0110100010010100
 $\xrightarrow{0110(+)}$ count of 1 = 3
 $(-) - (+) = (+)$; $\boxed{OF=1}$ $\therefore \boxed{PF=0}$

b) Possible. When three different segments cover a single memory location (overlap), that location will have three different logical addresses, while having a single physical address.

3) a) I)

CS:

Start: 35E90 H

End: 45E8F H

DS:

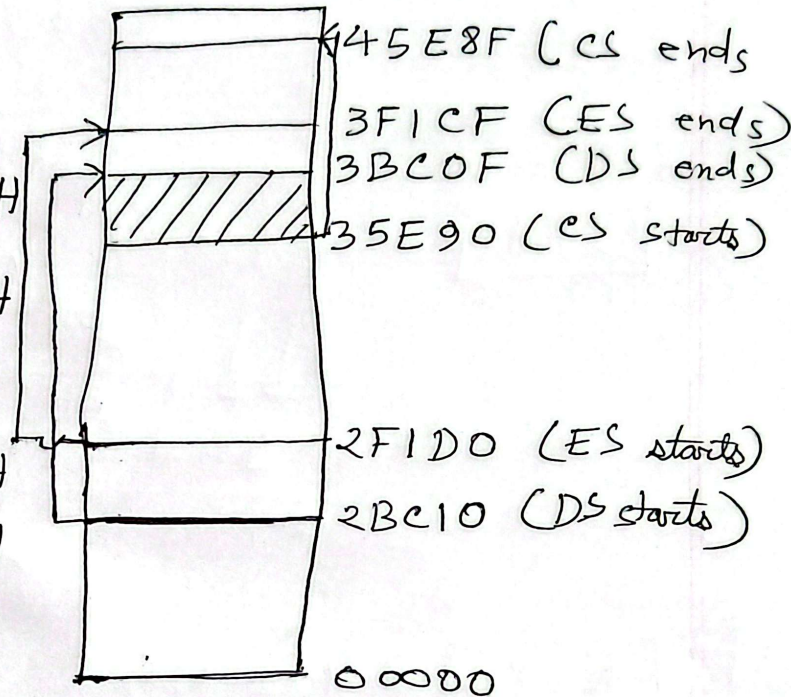
Start: 2BC10 H

End: 3BC0F H

ES:

Start: 2F1D0 H

End: 3F1CF H



∴ Total slots overlapped
by all 3 of them:

$$3BC0F - 35E90 + 1$$

$$= 5D80 H$$

$$= 23936 \text{ slots (in decimal)}$$

3(b)

SRAM

$$D.B = 16 \text{ bit}$$

64K addresses

DRAM

$$D.B = 8 \text{ bit}$$

$$A.B = x \text{ bit}$$

$$2^x \times 8 = 2 \times 64K \times 16$$

$$2^x \times 8 = 2 \times 2^6 \times 2^{10} \times 2$$

$$2^x = 2^{18}$$

$$\therefore x = 18 \text{ bit}$$

$$\text{SRAM capacity} = 64K \times 16 \text{ bit}$$

$$= 64K \times 2 \text{ byte}$$

$$= 128 \text{ Kb}$$

$$= 128 / 1024 \text{ Mb}$$

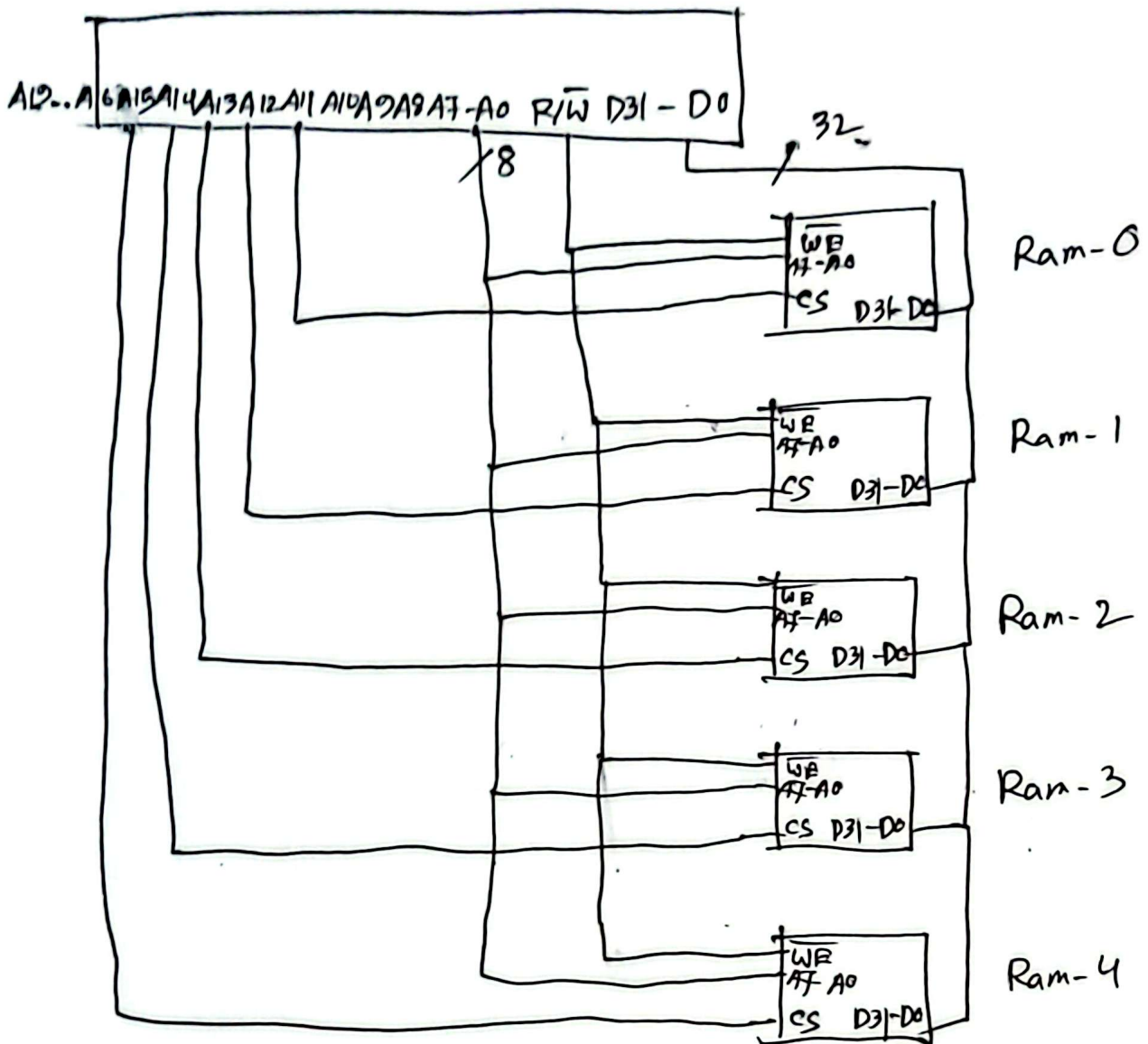
$$= 0.125 \text{ Mb}$$

$$\text{DRAM} \rightarrow 0.125 \times 2 = 0.25 \text{ Mb}$$

4 / Ram size = $(2^8 \times 32) \text{ bit} = 1 \text{ KB}$

$\therefore \text{No. of Rams} = \lceil 5 \text{ KB} / 1 \text{ KB} \rceil = 5$

S.A. = 00800H $\rightarrow$ 0000 0000 ^{All} 1000 0000 0000



3rd Ram (Ram-2)

S.A. = 02000H

4th Address = 02003H

6th " = 02005H

Last Ram (Ram-4)

S.A. = 08000H

4th Address = 08003H

6th " = 08005H

4(a) Last interrupt $79 = 4F H$

$$\therefore \text{last address} = 4F H \times 4H + 3H \\ = 013F H$$

(b) Denji's sensors interrupt no = $\frac{00138 H}{4H}$
 $= 4E$

$$4E < 4F \rightarrow \text{valid}$$

AWS " " " = $\frac{00193 - 3}{4H}$
 $= 64H$

$$64 > 4F \rightarrow \text{invalid}$$

(c) Denji's sensor

as lower number gets higher priority.